

Table 2. Measured boiling incipience superheats for all surfaces and fluids tested, along with calculated incipient bubble radii. Values in table are the maximum single-phase wall superheats measured and accuracy is based upon the magnitude of the input power increment.

Surface	HFE-7300 [*]		DI water [†]	
	$\Delta T_{w,i}$ [K]	$r_{b,i}$ [μm]	$\Delta T_{w,i}$ [K]	$r_{b,i}$ [μm]
Bare copper	22.5 ± 2.0	0.279 ± 0.026	9.7 ± 0.6	3.51 ± 0.22
Metallized CNT	12.6 ± 1.2	0.498 ± 0.049	4.8 ± 1.2	7.46 ± 1.89
Sintered copper	6.6 ± 2.3	1.07 ± 0.38	0.4 ± 0.1	88.6 ± 22.2
Sintered copper + metallized CNT	0.8 ± 0.2	8.27 ± 2.07	0.5 ± 0.2	79.2 ± 31.7

* Thermophysical property values estimated from [45]

† Thermophysical property values obtained from [46]